PDR: A paired diagnosis-and-repair framework for 8-bit object detectors under image corruption

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ABSTRACT

Quantized object detectors are deployed on edge accelerators, yet their robustness is commonly judged by comparing number formats on corrupted images alone. Such evaluations confound the clean-image gap with the corruption effect, ignore how each engine is built, and cannot locate or repair a failure. This paper proposes PDR, a paired diagnosis-and-repair framework that overcomes these limitations for executable 8-bit detectors. PDR feeds identical codec-matched bytes to INT8 and FP8 TensorRT engines, estimates the corruption-induced change in their accuracy gap by common-image bootstrap, localizes clean-level deficits by region decomposition and a weight/activation factorial, and repairs them by keeping only the implicated region in floating point, without retraining. Across nine YOLO11 blocks on KITTI, PASCAL VOC, and COCO, the format-level corruption effect reverses sign (four Holm-significant blocks of both signs), although FP8 leads in absolute AP throughout. In RetinaNet, PDR traces a 21.2-AP INT8 collapse to the activations consumed by the regression head. Selective precision outperforms the standard entropy-calibrated INT8 recipe by 13.85 clean-control AP (95% CI 12.76 to 15.00) on KITTI images unseen by a retrained checkpoint, recovering 99% of the gap to FP8. On the KITTI development set it exceeds the best of five calibration-only repairs by 19.8 AP, with 8–11% lower batch-1 engine latency than FP8. Corruption-aware calibration transfers inconsistently and can reduce held-out accuracy. On anchor-free FCOS, neither an INT8 deficit nor a repair gain appears.

1. Introduction

Object detectors on edge accelerators rely on post-training quantization (PTQ) to meet memory, energy, and latency budgets [1, 2]. Two 8-bit formats are compared here: 8-bit integer (INT8), with a mature integer inference path [3–5], and 8-bit floating point (FP8), which trades mantissa precision for exponent range [6, 7]. Deployed detectors also receive images degraded by noise, blur, weather, and compression, which harm deep visual models [8, 9] and detectors [10–12]. Selecting a quantized detector for such conditions is a deployment decision; in an analogous forecasting setting, model choices based on a single evaluation reference became unstable under plausible perturbations [13].

Current evaluation practice has three limitations. (L1) Quantized detectors are compared by format label, although an executable detector is defined by a *recipe*: calibration data and preprocessing, quantizer placement, excluded regions, graph transformations, and runtime settings [14]. (L2) Robustness is summarized either by the accuracy gap on corrupted images or by retention ratios such as relative performance under corruption (rPC) [10]. The first combines the clean-image difference with the change caused by corruption; the second measures proportional accuracy preservation within one model and does not estimate how corruption changes the gap between two engines. (L3) When a quantized detector fails, detector PTQ methods design new quantizers that optimize clean accuracy [15–17], and a recent study of degradation-aware INT8 calibration for YOLO detectors [18] did not test transfer to corruption families withheld from calibration. Evaluations seldom connect a measured deficit to the model region and operand responsible, and from there to a repair that needs no retraining.

This paper proposes PDR, a paired diagnosis-and-repair framework for executable 8-bit detectors that overcomes these limitations. Stage 1 feeds identical, codec-matched image bytes to the engines under comparison and estimates the corruption-induced change in their average-precision (AP) gap, ΔE, inside common-image bootstrap resamples, which removes the clean-gap confound (L2). Stage 2 localizes a level deficit through region decomposition and an operand factorial that quantizes the weights and activations of a region independently (L1). Stage 3 repairs the deficit by keeping only the implicated region in floating point, with checkpoint and calibration unchanged. Stage 4 verifies the repair on images unseen by a retrained checkpoint, on a second detector family, on independently pretrained checkpoints, and on engine latency. Calibration remedies are compared with the repair on entire corruption families withheld from calibration (L3).

Applied to TensorRT INT8 and FP8 engines of YOLO11, RetinaNet, and FCOS on KITTI, PASCAL VOC, and COCO, PDR yields three main results. First,

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the format-level corruption effect has no stable sign: across nine YOLO11 dataset×scale blocks, four blocks remain significant after Holm correction, with both signs, although FP8 leads INT8 in clean and corrupt-mean AP in every block. Second, the standard entropy-calibrated INT8 recipe for RetinaNet trails FP8 by 21.2 AP on KITTI, and PDR traces this deficit to the activations consumed by the regression head. Third, the selective-precision repair outperforms the standard INT8 recipe by 13.85 clean-control AP on KITTI images unseen by a retrained checkpoint, where its difference to FP8 is statistically indistinguishable from zero ($+0.21$ $[-0.48, +1.11]$). On the KITTI development set it exceeds the best of five calibration-only repairs by 19.8 clean-control AP, and its batch-1 engine latency is 8–11% below that of FP8 on the tested GPU. Corruption-aware calibration transfers inconsistently and can significantly reduce held-out accuracy.

The contributions are as follows.

- A paired, codec-controlled estimand with common-image bootstrap inference for executable quantized detectors. It removes the clean-image gap from corrupted-input comparisons and complements retention ratios with an absolute paired interaction.
- A localization procedure, region decomposition followed by an operand factorial, that attributes the RetinaNet INT8 deficit to regression-head activations; INT8 head weights cost at most about 0.2 AP.
- A retraining-free selective-precision repair that outperforms the standard INT8 recipe by 13.85 clean-control AP on a retrained-checkpoint holdout and 21.3 AP on KITTI validation, has FP8-minus-selective intervals that include zero on both KITTI sets, and has lower batch-1 engine latency than FP8.
- Evidence from two frozen calibration folds that corruption-aware calibration does not transfer reliably to held-out corruption families, together with decision rules for deploying quantized detectors (Section 6.2).

The framework and the experimental protocol are specified in Sections 3 and 4; Section 5 reports the results stage by stage, and Section 6 discusses mechanisms, decision rules, and limitations.

2. Related work

2.1. Post-training and mixed-precision quantization of detectors

PTQ estimates numerical ranges from representative data or data-free surrogates and reduces error through clipping, rounding, or local reconstruction [19–24]. Recent methods address hardware constraints and low-bit vision transformers [25–27]. Detectors are harder to quantize than classifiers because classification, localization, feature pyramids, and foreground–background imbalance [28] must remain calibrated together, and heads that decouple classification from regression [29, 30] give the two tasks separate numerical paths. Q-YOLO targets activation-distribution imbalance in YOLO PTQ [31], DetPTQ and Reg-PTQ address detectors through detection-loss-guided reconstruction and regression-specialized quantizers [15, 16], and InlierQ suppresses anomalous activations during calibration [17]. Reg-PTQ, the closest precedent, identifies regression branches as hard to quantize and proposes a dedicated quantizer [16]. These methods modify the quantizer and target clean accuracy. The methodological contribution of PDR is different: a clean-gap-adjusted paired comparison of executable engines is coupled with region and operand localization, diagnosis-guided precision selection, and verification, on an unmodified deployment toolchain.

Mixed-precision quantization assigns bit widths by layer sensitivity [4, 32, 33]; MPQ-YOLO assigns different precisions to the backbone and head of YOLO detectors with training [34], and quantization-aware training preserves detection accuracy at low precision at the cost of retraining [35]. In PDR, precision is assigned from measured regional deficits rather than from a sensitivity proxy: one region stays in floating point, and the checkpoint and calibration budget stay fixed.

2.2. INT8 versus FP8

Integer-only inference is well established [3–5], and FP8 now has practical inference and PTQ layouts [6, 7, 36]. The preferred 8-bit format varies by layer and architecture [14], so the datatype alone does not describe an engine. PDR records the full ModelOpt/TensorRT treatment and varies one recipe component at a time, which makes the recipe–format confound measurable.

2.3. Robustness of quantized and compressed models

Corruption benchmarks measure aggregate degradation of classifiers [8] and detectors [10, 11]. Detector-side remedies include image enhancement for adverse weather [12] and biologically inspired feature models [37]; for classifiers, training-time augmentation [38] and test-time adaptation [39–41] are common. The usual detector summary, $\mathrm{rPC} = \mathrm{mPC}/P_{\mathrm{clean}}$ [10], with mPC the mean performance under corruption and P_{clean} the clean performance, measures how much accuracy one model retains.

For quantized models, RobustMQ reports method- and architecture-dependent robustness of quantized classifiers under common corruptions [42], chaotic noise-injected quantization improves accuracy under adversarial attacks [43], and adversarial robustness under quantization has been studied separately [44]. For detectors, degradation-aware INT8 calibration has been evaluated across YOLO scales with inconsistent gains [18], and detection-discriminant reorganization combines compression with robustness [45]. More broadly, TTAQ couples perturbation-aware PTQ with adaptation under continual shift [46], and Recti-Q rectifies out-of-distribution features of quantized perception models [47].

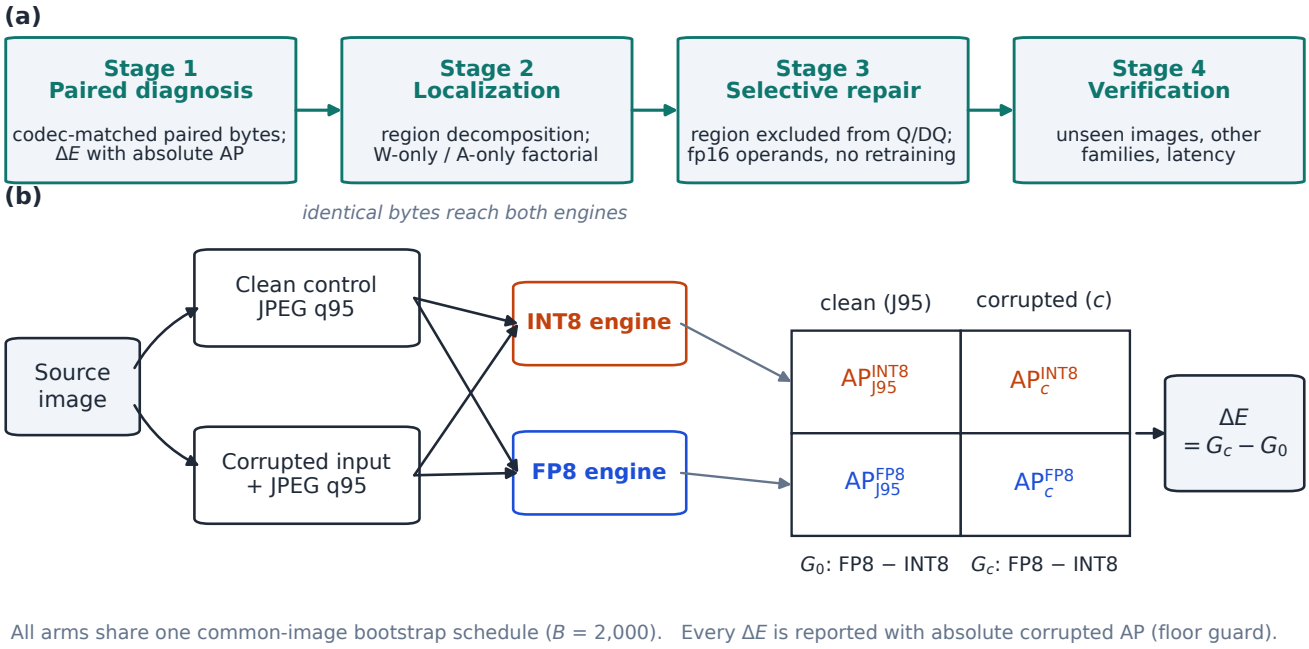


Figure 1: The PDR framework. (a) Four stages: paired diagnosis, localization, selective-precision repair, and verification. (b) Stage-1 protocol: each source image yields a clean control re-encoded at JPEG quality 95 (J95) and corrupted inputs that are also re-encoded at quality 95, so the INT8 and FP8 engines receive identical bytes. The four AP cells give the clean gap G_0 and the corrupted gap G_c, and $\Delta E = G_c - G_0$. The FP32 reference cancels from ΔE and is not shown.

These approaches address quantization, compression, or adaptation. In PDR, identical-input engine comparisons are coupled to regional diagnosis; the paired interaction is reported together with absolute AP, so that a gap shrinking near a shared failure floor is not read as robustness, and calibration transfer is tested on entire corruption families withheld from calibration.

3. Proposed PDR framework

3.1. Overview

PDR (Fig. 1, Algorithm 1) takes a trained detector, its candidate 8-bit recipes, and a corruption suite, and returns a per-region diagnosis, a repaired recipe, and its verified accuracy and latency. A *treatment* is the complete recorded configuration of an executable detector: checkpoint, Open Neural Network Exchange (ONNX) graph, quantize/dequantize (Q/DQ) placement, calibration list and preprocessing, excluded operators, TensorRT builder settings, runtime, and decoder. "INT8" and "FP8" therefore qualify treatments, not isolated datatypes; an *arm* is one treatment within an experiment. Algorithm 1 is a retrospective statement of the workflow, not an automated search: the comparisons of each stage were designed by hand, and Section 4.4 gives the order in which they were run and their inferential status.

3.2. Stage 1: paired corruption estimand

For dataset d, model m, treatment q, and condition c, $\mathrm{AP}_{d,m,q,c}$ denotes COCO-style AP@[.5:.95] in AP points, and J95 denotes the clean control, i.e., the source image encoded at JPEG quality 95 without corruption. The FP8–INT8 gaps on clean and corrupted inputs are

$$G_{d,m,x} = \mathrm{AP}_{d,m,\mathrm{FP8},x} - \mathrm{AP}_{d,m,\mathrm{INT8},x}, \quad x \in \{\mathrm{J95}, c\}, \tag{1}$$

and the primary interaction is

$$\Delta E_{d,m,c} = G_{d,m,c} - G_{d,m,\mathrm{J95}}. \tag{2}$$

A positive ΔE indicates that the FP8–INT8 gap widens under corruption, and a negative value that it narrows; the FP32 reference cancels. With $G_0 = G_{d,m,\mathrm{J95}}$ and G_{corr} the mean of $G_{d,m,c}$ over the 12 corruption–severity cells, the per-block estimand is $\Delta E = G_{\mathrm{corr}} - G_0$. The contrast has the algebra of a difference-in-differences interaction and is read descriptively, without a causal claim. ΔE is scale-sensitive: if both treatments retain the same fraction r of their clean AP, then $\Delta E = (r-1)G_0$, which is negative whenever $G_0 > 0$ and $r < 1$. Every ΔE is therefore reported with the clean gap and the absolute corrupted AP (the floor guard).

The same construction applies to two arms of one format (for example, INT8-selective minus INT8-matched), which isolates a single recipe component. Restricting the corrupted set to a subset S gives the calibration contrast

$$\Delta E_{A-B,S} = \left[\mathrm{AP}_A - \mathrm{AP}_B\right]_S - \left[\mathrm{AP}_A - \mathrm{AP}_B\right]_{\mathrm{J95}}, \tag{3}$$

denoted $\Delta E_{\text{held-out}}$ or $\Delta E_{\text{in-family}}$; for a calibration-variant-minus-baseline contrast, a positive value is a corruption-specific benefit. The *level deficit* of an arm is its AP shortfall to a comparator on clean inputs (G_0 when the comparator

Algorithm 1 Paired diagnosis and selective-precision repair (PDR), stated as a retrospective workflow of the analyses reported here

Require: FP32 detector; INT8 and FP8 recipes; evaluation images I; corruption cells C; calibration images disjoint from I; holdout images H
Ensure: Region diagnosis, repaired recipe, verified AP and latency
Match calibration preprocessing to inference byte for byte; build all arms from one ONNX graph
for all $i \in I$, $x \in \{\text{J95}\} \cup C$ **do**
    Encode i under x with terminal JPEG-95; run every arm on the identical bytes
end for
Stage 1: on one shared bootstrap schedule, estimate G_0, G_c, and $\Delta E = G_c - G_0$; report them with absolute AP
if the INT8 arm shows a clean-level deficit to FP8 **then**
    Stage 2: compare targeted arms that quantize only one region or all but one region (backbone, FPN, classification head, regression head)
    if one region reproduces the deficit when quantized alone and removes it when excluded **then**
        $r^\star \leftarrow$ that region
        Quantize the weights and activations of $r^\star$ separately (W-only, A-only) to identify the implicated operand
        Stage 3: exclude $r^\star$ from Q/DQ insertion; keep checkpoint and calibration fixed
    else
        Report the deficit as unlocalized; retain FP8 (fallback, not exercised here)
    end if
else
    Keep the INT8 recipe; report ΔE with absolute AP
end if
Stage 4: if a repaired recipe exists, re-estimate repaired−INT8 and FP8−repaired contrasts on H and on further detector families; measure batch-1 engine latency

is FP8); the interaction ΔE is what corruption adds. The RetinaNet INT8-matched arm, for example, has a large level deficit but a negative ΔE on KITTI.

3.3. Stage 2: region and operand localization

A level deficit is localized in two steps. *Region decomposition* builds INT8 arms that quantize one region only (backbone, feature pyramid network (FPN), classification head, or regression head, alone or combined) and arms that quantize everything except one region. Their J95 AP is compared with the FP32 and FP8 references, and recipe sweeps over preprocessing, range estimator, calibration budget, and Q/DQ placement test whether the split depends on these choices. The region $r^\star$ is the one whose quantization alone reproduces the deficit and whose exclusion alone removes it.

The *operand factorial* then quantizes only one operand class of $r^\star$: INT8 weights with fp16 activations (W-only) and fp16 weights with INT8 activations (A-only). Only the Q/DQ edges of $r^\star$ are rewired, and each rewired edge is checked against the selective graph. With the fully quantized and selective arms, this gives a 2×2 factorial over weights and activations, with interaction selective−W-only−A-only+matched. Rewiring acts on the consumer side, so "head activations" include the FPN feature maps that enter the head.

3.4. Stage 3: selective-precision repair

The repaired recipe, *INT8-selective*, excludes every node of $r^\star$ from Q/DQ insertion, so these convolutions have 16-bit floating-point (fp16) operands in the graph, while all other quantizers, the checkpoint, the calibration list, and the builder settings stay identical to the INT8 recipe. For RetinaNet, $r^\star$ is the `/head/regression_head/` subgraph; for FCOS, the same exclusion also covers the centerness output, which torchvision places in the regression head. The repair requires no retraining, no additional data, and no change to the TensorRT toolchain.

3.5. Stage 4: verification and statistical inference

Intervals describe uncertainty from the finite set of evaluation images, with checkpoints, engines, calibration, and corruptions fixed [48, 49]. Within each dataset×model block, one deterministic schedule of 2,000 image resamples [50] is shared by all arms and conditions, so every contrast is computed on common resamples. Point estimates are full-sample values; intervals are 2.5th–97.5th bootstrap percentiles, marked * when they exclude zero. Two-sided p-values use the plus-one rule $\tilde{p} = \min\{1, 2(b_{\min}+1)/(B+1)\}$, where $b_{\min}$ is the smaller of the numbers of draws ≤ 0 and ≥ 0, and are Holm-adjusted across the nine YOLO11 blocks [51]. Heterogeneity across blocks is tested with Cochran's Q [52, 53] and, because blocks of one dataset share images, with a generalized Q that uses the covariance from a rerun with a dataset-shared schedule. KITTI frames are correlated within drives, so KITTI intervals are also recomputed by resampling the 108 drives of the validation partition (B = 1,000; Supplement S9).

Verification has four parts: a KITTI holdout scored with a retrained checkpoint (Section 4.2), a second detector family (FCOS), off-the-shelf COCO-pretrained checkpoints, and deployment checks of batch-1 latency and engine-rebuild determinism (Supplement S12). Diagnostic arms are single runs without intervals.

4. Experimental setup

4.1. Detectors, engines, and corruptions

YOLO11n/m/x [54], RetinaNet with a ResNet-50 FPN [55, 56], and anchor-free FCOS [29, 57] are evaluated. All engines were built with TensorRT 11.1.0.106 [58] on an NVIDIA GeForce RTX 5090. Quantized graphs were produced by NVIDIA ModelOpt with entropy calibration [59, 60], except the max-calibration baselines of Section 4.3; study-built registries record ModelOpt 0.45.0, and the older YOLO11 registries do not record its version. The YOLO11 engines predate the frozen analysis and were built by `trtexec` with TensorRT defaults (TensorFloat-32, TF32, enabled); the RetinaNet and FCOS quantized arms

were built with the TensorRT Python builder and TF32 disabled. Within each family, all quantized arms entering a contrast share one TF32 setting. Paired and calibration arms use 512 deterministically selected training images, disjoint from the evaluation images.

The RetinaNet arms are **INT8-legacy** (the historical recipe, whose calibration preprocessing differs from inference; diagnostic only), **INT8-matched** (the standard entropy INT8 recipe with calibration preprocessing identical to inference), **INT8-selective** (Section 3.4), the calibration baselines of Section 4.3, **FP8-matched**, and **FP32**. Supplement S4 gives the full arm registry and terminology.

Four synthetic corruptions are generated deterministically at three severities (labels 1/3/5): Gaussian noise ($\sigma \in \{8, 20, 40\}$ on the 0–255 scale), linear motion blur (kernel lengths $\{5, 13, 23\}$, random angle), fog (strength–radius pairs $(0.18, 6)$, $(0.38, 12)$, $(0.62, 20)$), and JPEG compression (qualities $\{70, 40, 15\}$). Every output is re-encoded at JPEG quality 95, so clean controls and corrupted inputs share the final codec, and a SHA-256-derived seed binds dataset, image, corruption, and severity. Inference uses a 640-pixel letterbox and family-specific decoders: YOLO11 uses a confidence threshold of 0.001 and class-wise non-maximum suppression (NMS) at an intersection-over-union (IoU) of 0.7; RetinaNet uses 0.05 and batched NMS at IoU 0.5; both keep at most 300 detections. AP follows the COCO protocol (`pycocotools`, `maxDets`=100).

4.2. Datasets and checkpoints

Three evaluation sets are used: KITTI validation (1,496 images) [61], PASCAL VOC2007 test (4,952 images) [62], and a fixed 2,000-image subset of COCO val2017 [63] (seed 20260924). VOC and KITTI annotations were converted from Ultralytics YOLO labels to COCO format; VOC objects marked difficult are omitted, and KITTI keeps eight categories and drops `DontCare` regions. KITTI and VOC YOLO11 models were fine-tuned from Ultralytics weights; RetinaNet and FCOS ResNet-50-FPN models were fine-tuned for 100 epochs at 640-pixel input (seed 20260807). Checkpoints were selected on the evaluation partition, so absolute KITTI and VOC AP is optimistic; whether selection affects the two arms of a contrast differently is unknown. COCO YOLO11 uses the official checkpoint. Each block covers original clean images, the codec control, and 12 corruption cells (14 conditions per arm; 13 for the older RetinaNet FP32 and legacy arms, which predate the codec control). All checkpoint, split, and training manifests are hash-bound (Supplement S4).

The KITTI evaluation set is the 1,496-image Ultralytics validation partition of the 7,481-image training set, and all KITTI checkpoints of the main experiments were trained on the complementary 5,985 images. For an unseen-image test, RetinaNet was retrained with the same profile on 4,788 of these images and selected on the 1,496-image set; the remaining 1,197 images form the holdout. For the retrained checkpoint, the holdout images were excluded from training, checkpoint selection, and calibration, but the recipes were developed earlier on checkpoints that had seen them. KITTI is split by frame, so drives cross every partition boundary: 3% of the 512 calibration images (13/512) have an identical-hash evaluation frame under a 16×16 average hash, and 12% have a match within distance 5. The holdout is therefore image-disjoint but not sequence-disjoint; it is analyzed with both the image and the drive-clustered bootstrap. The KITTI validation set was used while the selective recipe was developed, so KITTI-validation selective estimates are development-contaminated; VOC, the COCO-pretrained replication, and the retrained holdout are cleaner checks.

4.3. Baseline repairs

Selective precision is compared with five retraining-free repairs that change only calibration.

Corruption-aware calibration. The calibration baseline tests one hypothesis under decision rules frozen before any of its runs: *if INT8-matched corrupted-input accuracy is limited because calibration images never show corrupted activation ranges, calibration on corrupted images should recover AP on held-out corruptions.* Each of the 512 calibration images is assigned by a hash of its path to one of six in-family cells and corrupted with the evaluation parameters and JPEG-95 encoding. Fold 1 calibrates on Gaussian noise and JPEG and holds out fog and motion blur; fold 2 swaps the roles (75–102 images per cell; Fig. 2). The full tier re-quantizes INT8-matched, and the selective tier re-quantizes INT8-selective. The fold-1 primary criterion requires the held-out corrupt mean of the full-tier arm to exceed that of INT8-matched with an interval above zero; the fold-2 replication rule is registered two-sidedly and is read through the directional fold-1 criterion it cites. Fold 2 also requires $\Delta E_{\text{held-out}} > 0$ against INT8-matched and against the codec control. Supplement S6 lists every rule and its registration status. In-family gains alone indicate calibration fitting, not transfer, and all rules are reported regardless of outcome.

Codec-matched calibration. Corrupt-calibration images carry the JPEG-95 encoding of the evaluation images, whereas the standard calibration list uses the original files. The INT8-q95calib controls, calibrated on the same 512 clean images re-encoded at quality 95, separate codec adaptation from corruption coverage. They are post hoc for fold 1 and pre-specified for fold 2.

Max calibration. INT8-matched and INT8-selective were rebuilt with max instead of entropy calibration. In a more local variant, the scale of every activation quantizer consumed by the regression head was set to the calibration-set maximum of its tensor, with all other scales unchanged. Five of these quantizers sit on FPN inputs shared with the classification head, so the patch is local to the regression tower but not at the shared boundary. The patch is scripted (scale = $\mathrm{fp16}(|x|_{\max}/127)$); it reproduces the archived KITTI graph byte for byte and the nodes and initializers of the VOC graph exactly. Both variants are single builds reported as point estimates (Supplement S14).

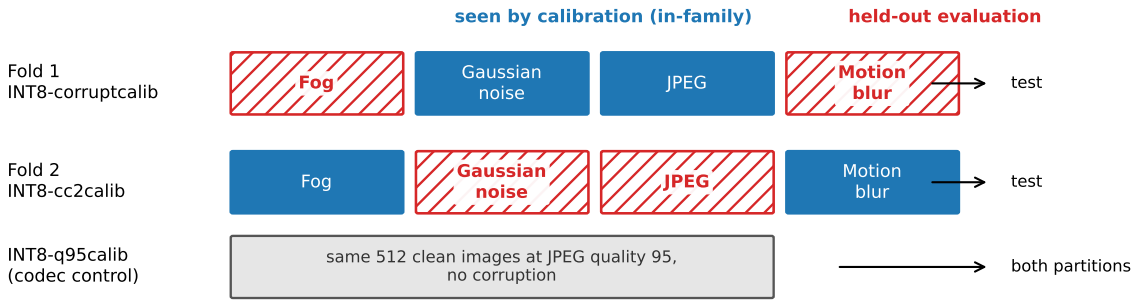


Figure 2: Corruption-aware calibration folds and the codec-matched control arm. Filled blue cells are families seen by entropy calibration (the in-family partition); hatched red cells are held out: their named corruption settings never enter calibration, although every calibration image carries the JPEG-95 encoding. Fold 1 (INT8-corruptcalib) calibrates Gaussian noise and JPEG and is tested on fog and motion blur; fold 2 (INT8-cc2calib) swaps the roles. INT8-q95calib sees only the same 512 clean images re-encoded at JPEG quality 95 and is evaluated on both partitions. Selective-tier arms (sel-corruptcalib, sel-cc2calib, sel-q95calib) share each fold's split.

4.4. Evidence status

The analyses were run in a different order from the stages. Diagnostic single-run decompositions on KITTI identified the regression head and the selective recipe first; the estimands, condition schedule, and bootstrap plan for evaluating the already chosen YOLO11 and RetinaNet arms were then frozen; the operand factorial and the retrained holdout followed. The freeze therefore covers the Stage-1 and Stage-3 evaluation, not the discovery of the repair or the four-stage workflow, and the YOLO11 engines predate it. The calibration folds were frozen before their runs. The region decomposition is diagnostic: single development-set builds that guided the design. The operand factorial, the FCOS and COCO-pretrained replications, the retrained holdout, the max-calibration counterfactuals, and the activation, latency, and rebuild audits were specified after the paired freeze and are reported regardless of outcome. Holm adjustment covers the nine YOLO11 blocks, a family chosen at analysis time; other intervals are unadjusted. An earlier 144-cell exploratory YOLO11 analysis motivated the frozen design and is reported only in Supplement S10. Supplement S10 tabulates the inferential status of every layer.

5. Results

5.1. Stage 1: the format-level corruption interaction has no stable sign

Table 1 and Fig. 3 report the nine YOLO11 blocks. FP8 exceeds INT8 in clean-control AP in every block (G_0 from +0.68 to +3.97 AP) and in the 12-cell corrupt mean, although a few individual cells reverse the order. The change in this gap under corruption, however, has no stable sign: ΔE is positive on VOC/YOLO11n (+0.31 [+0.07, +0.59]) and COCO-2k/YOLO11m (+0.60 [+0.27, +0.97]), negative on VOC/YOLO11m (−0.57 [−0.80, −0.34]), KITTI/YOLO11m (−0.68 [−1.32, −0.05]), KITTI/YOLO11x (−0.69 [−1.13, −0.33]), and COCO-2k/YOLO11x (−0.53 [−0.93, −0.18]), and compatible with zero elsewhere (unadjusted). On VOC alone, the sign changes with model scale (+0.31, −0.57, and +0.14 for n, m, and x).

Cochran's test rejects a common ΔE ($Q = 63.3$, 8 df, $p \approx 1 \times 10^{-10}$; $I^2 = 87\%$; $\tau = 0.44$ AP). Accounting for the shared images within a dataset through the covariance of a shared-schedule rerun ($B = 1{,}000$) leaves the result unchanged (generalized $Q = 67.1$, $p \approx 2 \times 10^{-11}$). Heterogeneity also holds within VOC ($Q = 31.0$, $p \approx 2 \times 10^{-7}$) and COCO ($Q = 18.9$, $p \approx 8 \times 10^{-5}$), where only model scale varies, but not within KITTI ($Q = 3.3$, $p = 0.19$). After Holm adjustment, 4 blocks remain significant with *both* signs: COCO-2k/YOLO11m is positive, and KITTI/YOLO11x, VOC/YOLO11m, and COCO-2k/YOLO11x are negative. The DerSimonian–Laird mean of −0.18 AP ([−0.50, +0.13]) is only a descriptive average.

Heterogeneity also appears across corruption families within blocks (Supplement S9); on VOC/YOLO11x, for instance, JPEG contributes −1.07 AP whereas Gaussian noise and motion blur contribute +0.69 and +0.67 AP. Removing the saturated motion-blur-s5 cell preserves the sign of every Holm-significant block ($Q \approx 60.2$). Under the drive-clustered bootstrap, KITTI/YOLO11x remains negative (−0.69 [−1.71, −0.33]), whereas KITTI/YOLO11m becomes compatible with zero (−0.68 [−0.90, +0.15]). The codec control behaves as intended: J95 AP stays within 0.3 AP of original clean AP for every arm, and within-model retention ranges from 0.68 to 0.79, with INT8 and FP8 differing by at most 0.02. Retention-based summaries would therefore rate the two formats as equally robust, whereas the paired estimand reveals significant interactions of both signs. With the recipe fixed, the interaction depends on dataset and model, and no format-level rule predicts it.

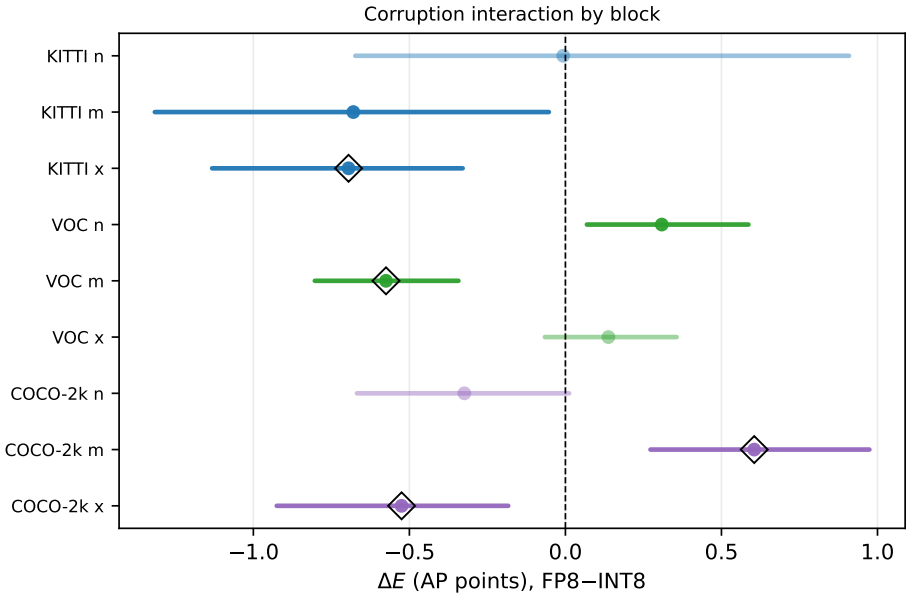


Figure 3: Paired corruption interaction ΔE (FP8−INT8) for the nine YOLO11 blocks with 95% common-image bootstrap intervals. Opaque markers and intervals: unadjusted interval excludes zero; faded: interval includes zero. Open diamonds: significant after Holm adjustment across the nine blocks ($p_{\text{Holm}} < 0.05$). The sign changes across dataset×model blocks, so no single format-level effect summarizes the interaction.

Table 1
Stage-1 paired contrasts, FP8 minus INT8, per YOLO11 dataset×model block. FP8/INT8 corr. AP: absolute 12-cell corrupt-mean AP (floor guard). G_0: clean-control (J95) gap. G_{corr}: corrupt-mean gap, computed before the absolute columns are rounded. $\Delta E = G_{\text{corr}} - G_0$ with paired-bootstrap interval; * interval excludes zero. p_{Holm}: two-sided bootstrap p-value, Holm-adjusted across the nine blocks, with the plus-one rule at B=2000; the smallest possible raw and adjusted values are $2/2001 \approx 0.001$ and $9\times2/2001 \approx 0.009$. Bold: blocks significant after Holm adjustment.

Dataset	Model	FP8 corr. AP	INT8 corr. AP	G_0	G_{corr}	ΔE	95% interval	p_{Holm}
KITTI	YOLO11n	44.1	41.9	+2.14	+2.13	-0.01	[−0.67, +0.91]	0.915
KITTI	YOLO11m	50.4	47.1	+3.97	+3.29	-0.68*	[−1.32, −0.05]	0.128
KITTI	YOLO11x	51.0	50.3	+1.44	+0.74	**-0.69***	**[−1.13, −0.33]**	**0.016**
VOC	YOLO11n	44.8	43.8	+0.68	+0.99	+0.31*	[+0.07, +0.59]	0.080
VOC	YOLO11m	51.8	50.7	+1.70	+1.12	**-0.57***	**[−0.80, −0.34]**	**0.009**
VOC	YOLO11x	54.3	53.3	+0.91	+1.05	+0.14	[−0.07, +0.36]	0.356
COCO-2k	YOLO11n	28.2	27.4	+1.10	+0.78	-0.32	[−0.67, +0.01]	0.171
COCO-2k	YOLO11m	38.0	36.5	+0.89	+1.49	**+0.60***	**[+0.27, +0.97]**	**0.016**
COCO-2k	YOLO11x	42.4	41.9	+1.00	+0.47	**-0.53***	**[−0.93, −0.18]**	**0.024**

5.2. Stage 2: localization in RetinaNet

Contract matching. About half of the historical INT8 collapse is caused by mismatched calibration preprocessing. In a controlled diagnostic that changes only the preprocessing (TF32 disabled, 128 calibration images), KITTI J95 AP rises from 6.72 to 27.91, recovering 21.2 of the 41.8-point gap to the diagnostic FP8 build. After this repair, INT8-matched still trails FP8-matched by 21.2 AP on KITTI but only 2.9 AP on VOC (original clean images; Supplement Table S4 and Fig. 4). The interaction ΔE is -4.69 $[-5.49, -3.86]$ on KITTI and -0.26 $[-0.58, +0.06]$ on VOC. The KITTI narrowing follows from the scale property of ΔE: INT8-matched starts about 21 AP lower and both arms retain similar fractions of their clean AP (0.72 versus 0.75). The deficit is therefore a level shift, which Stage 2 localizes.

Region decomposition. The diagnostic arms on KITTI place the collapse in the regression head (Table 2; single runs). Quantizing only the classification head preserves accuracy (50.57), quantizing only the regression head reproduces the collapse (16.41), and quantizing everything except the regression head gives 49.50, within 0.94 AP of the diagnostic FP32 build (50.44). Across all eleven INT8 arms, every arm that quantizes the regression head scores 6.72–28.74 AP, regardless of preprocessing, estimator, budget, or placement, and every arm that does not scores 49.14–50.57 AP. Graph position does not explain the split, because the equally terminal classification head shows no measurable loss; nor does the number of quantized sites, because the restricted Conv+Add placement (15.75) quantizes fewer sites than full INT8 yet does worse.

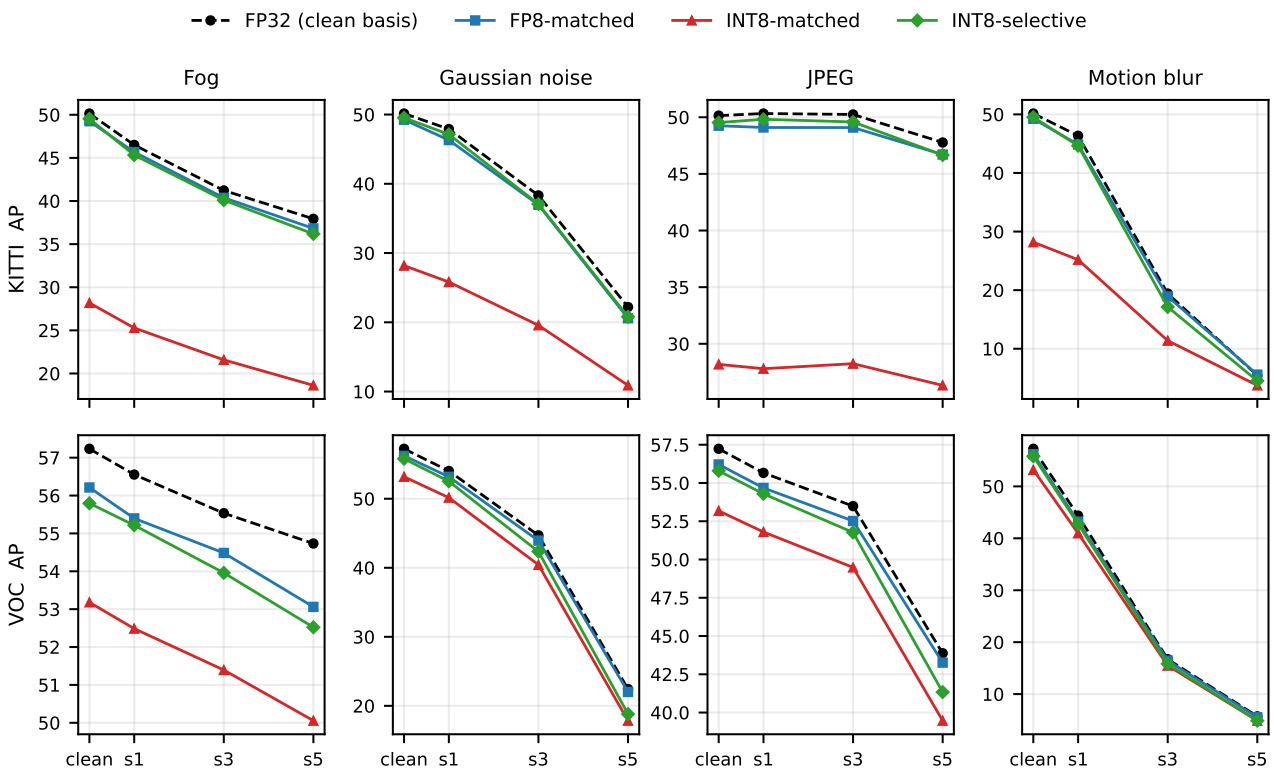


Figure 4: RetinaNet AP by corruption family and severity for the four matched-protocol arms on KITTI (top) and VOC (bottom). The "clean" point is the J95 codec control for the matched arms and the original clean image for FP32 (dashed), which predates the codec control. INT8-matched lies far below the other arms on KITTI and somewhat below them on most VOC cells. INT8-selective tracks FP8-matched on most families; on KITTI its residual grows significantly under motion blur. All four arms collapse at motion-blur severity 5.

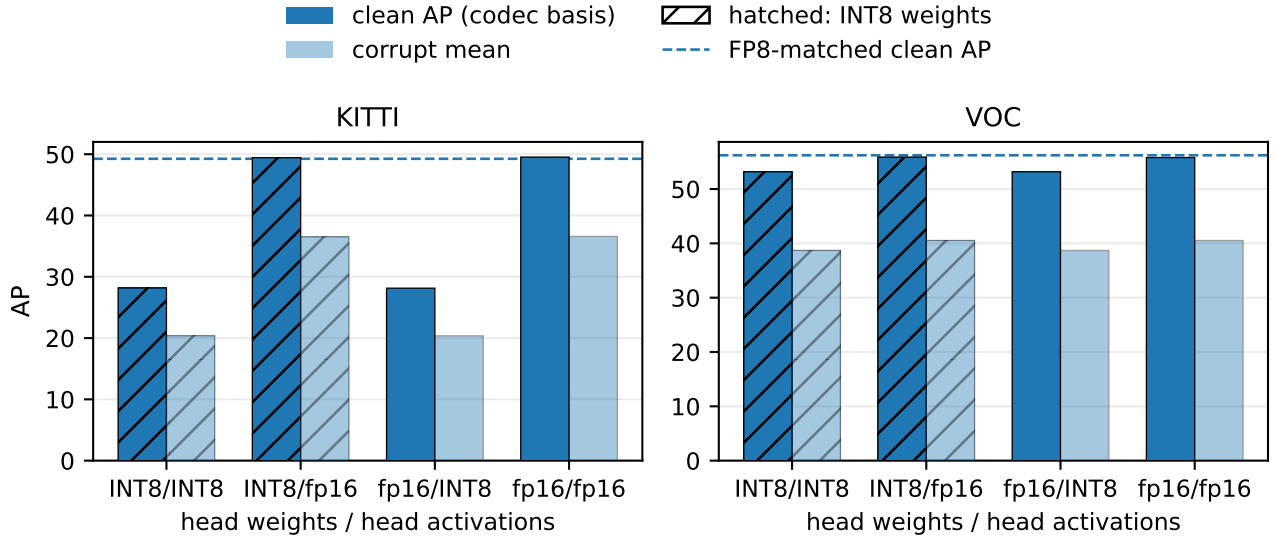


Figure 5: Stage-2 operand factorial inside the RetinaNet regression head (post-hoc arms, frozen paired conditions). Head weights and head activations are set to INT8 or fp16 independently. Solid bars: codec-control clean AP; translucent bars: 12-cell corrupt mean. With INT8 weights and fp16 activations, AP matches the selective arm within 0.1 on both datasets; with fp16 weights and INT8 activations, it matches the fully quantized arm. The deficit follows activation quantization.

Table 2
Stage-2 region decomposition on KITTI (diagnostic single runs; J95 AP; 512 calibration images unless stated). Every arm that quantizes the regression head collapses, and every arm that does not stays near FP32. The "Matched preprocessing, 512 img" row uses the same graph as the paired INT8-matched arm. Bold: the two arms that isolate the regression head.

INT8 arm	Reg. head INT8	J95 AP
Legacy preprocessing, 128 img	yes	6.72
Matched preprocessing, 128 img	yes	27.91
Matched preprocessing, 512 img	yes	28.18
Max estimator, 128 img	yes	23.94
Conv+Add sites only, 128 img	yes	15.75
FP8 compute sites (shared mask)	yes	28.74
Both heads only	yes	17.09
Regression head only	yes	**16.41**
Classification head only	no	50.57
Backbone + FPN only	no	49.14
All except regression head	no	**49.50**
FP8 matched, 128 img (reference)	–	48.50
FP32 (reference)	–	50.44

Operand factorial. The factorial (Fig. 5, Table 3) attributes the deficit to activation quantization. Restoring only the head *activations* to fp16 raises J95 AP to 49.44 on KITTI and 55.88 on VOC, recovering 100% and 103% of the selective gain; the remaining difference to full exclusion is below 0.2 AP (+0.09 [+0.01, +0.17] on KITTI, −0.09 [−0.20, +0.02] on VOC). Restoring only the head *weights* leaves contrasts compatible with zero (A-only minus matched: −0.04 [−0.20, +0.11] on KITTI, +0.01 [−0.08, +0.10] on VOC). INT8 head weights thus cost at most about 0.2 AP; the KITTI selective-minus-W-only interval just excludes zero. The interaction is compatible with zero on clean inputs (+0.14 [−0.04, +0.30] on KITTI, −0.09 [−0.23, +0.05] on VOC) and small under corruption (+0.07 [+0.01, +0.14] on KITTI; +0.01 [−0.03, +0.05] on VOC). Stage 2 therefore identifies the regression head as $r^\star$ and activation quantization as the implicated operand.

5.3. Stage 3: selective precision versus alternative repairs

Table 4 compares selective precision with the five calibration-based repairs. Selective precision raises KITTI J95 AP by 21.3 points over the standard INT8 recipe and exceeds the best calibration-based repair by 19.8 AP on clean and 14.9 AP on corrupted inputs; the best calibration gain is only +1.56 AP. On VOC, where the deficit is smaller, the margins are 1.6 and 0.6 AP. Under the frozen protocol, INT8-selective is compatible with FP8-matched on KITTI for both the clean control (−0.28 [−1.11, +0.56], FP8 minus selective) and the corrupt mean (+0.17 [−0.07, +0.40]); the clean interval spans more than one AP point, so this bounds the difference without proving equivalence. Relative to the clean gap, the residual widens under motion blur (+1.30 [+0.51, +2.04]). On VOC a small residual remains (+0.42 [+0.13, +0.74] clean, +0.96 [+0.83, +1.09] corrupt mean), so the regression head is the dominant but not the only source of the deficit there.

Corruption-aware calibration. Table 5 summarizes the two folds on two bases: the raw held-out AP change and the corruption-specific component $\Delta E_{\text{held-out}}$ (Eq. (3)); Supplement S7 gives all contrasts. On the raw basis, the primary criterion is met on KITTI fold 1 and VOC fold 2 and is null on VOC fold 1. On KITTI fold 2, raw AP drops significantly relative to INT8-matched and INT8-q95calib, including on the calibrated families, so the criterion fails. Where the criterion is met, the gain is small: on KITTI fold 1 it closes less than a tenth of the gap to FP8-matched (−13.34 [−14.13, −12.57] remains). On the interaction basis, only VOC fold 2 shows a significantly positive corruption-specific component (+0.62 [+0.30, +0.96]; +1.06 [+0.76, +1.38] against the codec anchor), VOC fold 1 is significantly negative (−0.71 [−1.02, −0.39]), and both KITTI cells are compatible with zero. Calibration on JPEG-95 images alone produces no detected change in clean AP (−0.34 [−0.93, +0.35] on KITTI, +0.20 [−0.07, +0.48] on VOC), but its interaction effects are as large as the corruption-specific components, so the codec-anchored contrasts are used to judge whether gains are corruption-specific. In the selective tier, the null-seeking fold-1 criterion (held-out residual to FP8-matched compatible with zero) fails on both datasets (−0.51 [−0.66, −0.38] on VOC, −1.30 [−1.61, −1.00] on KITTI).

Max calibration. Graph-wide max calibration covers the full calibration range yet deepens the KITTI deficit (14.02 versus 28.29 AP on original clean images) and destroys the VOC classifier's score distribution (Supplement S14). Restricting max calibration to the quantizers consumed by the regression head keeps VOC functional but does not repair the deficit (27.15 and 52.32 AP versus 28.29 and 53.30). No calibration-only repair approaches selective precision, and the deficit is not an artifact of entropy calibration. Combining selective precision with corruption-aware calibration can add a little more (Supplement Table S4), so the comparison concerns calibration as a substitute for the repair, not as a complement.

Severity wall. At motion-blur severity 5 every matched-protocol arm, FP32 included, falls below 6 AP on both datasets (KITTI: FP32 5.45, FP8-matched 5.59, INT8-matched 3.75, selective 4.52). This shared failure compresses absolute AP gaps in that cell; removing it changes no frozen-criterion verdict (Supplement S9).

5.4. Stage 4: verification on unseen images and other detectors

Retrained KITTI holdout. The pattern replicates on the 1,197 KITTI images excluded from training, selection, and calibration of the retrained checkpoint (Table 6; Supplement S15). Earlier checkpoints used during recipe development had been trained on these images, so this is a post-hoc, checkpoint-specific check. INT8-matched trails FP8-matched by +14.06 [+12.85, +15.35] AP on J95 inputs. Selective

Table 3
Operand decomposition of the regression-head exclusion (post-hoc arms; 95% paired-bootstrap intervals on the J95 basis relative to the INT8/INT8 matched arm). "J95 vs W8A8": clean-level contrast to the matched arm. Unquantized operands run in fp16. Retention: corrupt-mean AP over J95 AP. *: unadjusted 95% interval excludes zero. Bold: the W-only and A-only contrasts that separate the two operands.

Dataset	Head weights	Head acts.	J95	Corrupt mean	Retention	J95 vs W8A8
KITTI	INT8	INT8	28.18	20.37	0.72	–
	INT8	fp16	49.44	36.52	0.74	**+21.25*[+19.88, +22.69]**
	fp16	INT8	28.14	20.36	0.72	**−0.04 [−0.20, +0.11]**
	fp16	fp16	49.53	36.58	0.74	+21.35*[+19.95, +22.81]
VOC	INT8	INT8	53.18	38.71	0.73	–
	INT8	fp16	55.88	40.53	0.73	**+2.70*[+2.46, +2.96]**
	fp16	INT8	53.19	38.68	0.73	**+0.01 [−0.08, +0.10]**
	fp16	fp16	55.79	40.52	0.73	+2.61*[+2.37, +2.88]

Table 4
Retraining-free repairs of the RetinaNet INT8 deficit under the frozen paired conditions. J95: codec-control clean AP; Δ: J95 AP minus standard INT8; Corrupt: equal-weight mean over the 12 corruption–severity cells. All entries are full-sample plug-in estimates, and the margins between rows are descriptive; Section 5.3 gives the paired intervals of the bootstrap arms. Calibration rows change only the calibration images or the range estimator; max-calibration rows are single builds. KITTI selective estimates are development-contaminated (Section 4.2); Section 5.4 reports the unseen-image holdout. [a]Original clean image; the FP32 arm predates the codec control, so no Δ is given. Bold: the proposed repair.

		KITTI			VOC		
Strategy	Changed component	J95	Δ	Corrupt	J95	Δ	Corrupt
Standard INT8 (entropy, matched)	–	28.18	–	20.37	53.18	–	38.71
Codec-matched calibration	calibration images	27.85	−0.34	20.42	53.39	+0.20	38.54
Corruption-aware calibration, fold 1	calibration images	29.74	+1.56	21.71	53.91	+0.73	39.35
Corruption-aware calibration, fold 2	calibration images	25.47	−2.71	18.68	54.22	+1.04	39.97
Max calibration, whole graph	range estimator	14.18	−14.01	7.82	0.00	−53.18	0.00
Max calibration, reg.-head quantizers	range estimator	27.43	−0.75	19.58	52.16	−1.02	37.93
Selective precision (proposed)	**quantized region**	**49.53**	+21.35	**36.58**	**55.79**	+2.61	**40.52**
FP8 (entropy, matched)	number format	49.25	+21.07	36.75	56.21	+3.03	41.48
FP32 reference	no quantization	50.15[a]	–	37.80	57.23[a]	–	42.32

Table 5
Corruption-aware calibration per fold×dataset cell, full tier (corrupt-calibration arm minus INT8-matched; AP with 95% paired intervals, unadjusted). "Held-out": raw AP change on the held-out families; $\Delta E_{\text{held-out}}$ subtracts the clean-control shift. Verdicts use the directional reading of the frozen criterion (Supplement S6).

Cell	Held-out	$\Delta E_{\text{held-out}}$	Verdict
KITTI fold 1	+1.07 [+0.87, +1.26]	−0.50 [−1.01, +0.03]	met
VOC fold 1	+0.02 [−0.13, +0.16]	−0.71 [−1.02, −0.39]	null
KITTI fold 2	−2.02 [−2.31, −1.73]	+0.69 [−0.10, +1.37]	harm
VOC fold 2	+1.66 [+1.52, +1.83]	+0.62 [+0.30, +0.96]	met, specific

Table 6
Retrained KITTI holdout (1,197 images unseen by the retrained checkpoint; 95% image-bootstrap intervals, unadjusted). Bold: the selective repair.

Contrast	J95	Corrupt mean
FP8 − INT8	+14.06 [+12.85, +15.35]	+10.29 [+9.49, +11.16]
Selective − INT8	**+13.85 [+12.76, +15.00]**	**+10.41 [+9.63, +11.23]**
W-only − INT8	+13.90 [+12.80, +15.01]	+10.40 [+9.62, +11.21]
A-only − INT8	−0.02 [−0.22, +0.18]	−0.09 [−0.16, −0.04]
FP8 − selective	+0.21 [−0.48, +1.11]	−0.12 [−0.33, +0.10]

precision recovers +13.85 [+12.76, +15.00] AP, 99% of this gap, leaving FP8 minus selective at +0.21 [−0.48, +1.11]. Restoring only the head activations recovers the same amount (W-only minus matched +13.90 [+12.80, +15.01]), whereas restoring only the head weights does not (A-only minus matched −0.02 [−0.22, +0.18]). The deficit and its repair also hold under the drive-clustered bootstrap (+14.06 [+10.92, +16.60] and +13.85 [+10.98, +16.28]).

FCOS. On anchor-free FCOS the INT8-matched arm shows no deficit of the RetinaNet kind (Table 7): J95 AP is 47.53 on KITTI and 49.02 on VOC, against 47.60 and 49.22 for FP32, and FP8 does not exceed INT8 (FP8−INT8 −0.47 [−0.990, −0.001] on KITTI, −0.14 [−0.33, +0.07] on VOC). Excluding the regression head changes nothing measurable (selective minus matched 0.00 [−0.15, +0.12] and +0.01 [−0.04, +0.08]), and the FP8–INT8 interaction is compatible with zero (−0.04 [−0.44, +0.40], +0.07 [−0.13, +0.25]). Stage 1 thus finds no INT8 deficit relative to FP8, and the transferred regression-head exclusion yields no detectable gain, so the plain INT8 recipe is retained; deficits in other regions were not searched for. FCOS also differs in anchor density and centerness scoring, and its selective arm additionally leaves centerness unquantized, so this result supports recipe

Table 7
Cross-architecture check (post-hoc arms): the RetinaNet regression-head deficit does not appear on FCOS. Paired-bootstrap contrasts at 95% under the frozen conditions. Each detector is a separate bootstrap block; no cross-detector contrast is estimated. *: unadjusted 95% interval excludes zero. Bold: the head-exclusion effect (selective minus INT8-matched).

Detector	Dataset	FP8−INT8 J95	FP8−INT8 ΔE	Sel−INT8 J95	Sel−INT8 ΔE	FP8−Sel J95
RetinaNet	KITTI	+21.07*[+19.57, +22.61]	−4.69*[−5.49, −3.86]	**+21.35*[+19.95, +22.81]**	−5.14*[−5.94, −4.34]	−0.28 [−1.11, +0.56]
FCOS	KITTI	−0.47*[−0.990, −0.001]	−0.04 [−0.44, +0.40]	**0.00 [−0.15, +0.12]**	0.00 [−0.13, +0.15]	−0.47 [−0.99, +0.05]
RetinaNet	VOC	+3.03*[+2.66, +3.41]	−0.26 [−0.58, +0.06]	**+2.61*[+2.37, +2.88]**	−0.80*[−0.98, −0.61]	+0.42*[+0.13, +0.74]
FCOS	VOC	−0.14 [−0.33, +0.07]	+0.07 [−0.13, +0.25]	**+0.01 [−0.04, +0.08]**	−0.03 [−0.09, +0.03]	−0.16 [−0.36, +0.05]

Table 8
Replication on pretrained checkpoints, COCO val2017 (paired 2,000-image subset; 95% paired-bootstrap intervals, unadjusted). Arm rows give J95 clean AP and corrupt-mean AP. Contrast rows give the difference on the J95 basis (column 3) and the interaction ΔE (column 4). *: interval excludes zero.

Detector	Arm / contrast	J95-basis AP	Corrupt-mean / ΔE
RetinaNet	FP32	38.30	31.18
	FP8-matched	37.84	30.58
	INT8-matched	34.70	27.97
	INT8-selective	37.27	29.91
	FP8−INT8	+3.14*[+2.78, +3.57]	−0.53*[−0.90, −0.14]
	Sel−INT8	+2.58*[+2.33, +2.85]	−0.64*[−0.84, −0.42]
	FP8−Sel	+0.56*[+0.22, +0.96]	+0.11 [−0.28, +0.48]
FCOS	FP32	34.32	27.61
	FP8-matched	34.24	27.51
	INT8-matched	34.19	27.44
	INT8-selective	34.20	27.45
	FP8−INT8	+0.05 [−0.22, +0.27]	+0.02 [−0.20, +0.28]
	Sel−INT8	+0.01 [−0.06, +0.09]	0.00 [−0.07, +0.07]
	FP8−Sel	+0.04 [−0.21, +0.24]	+0.02 [−0.19, +0.27]

dependence rather than a claim about anchor-based heads in general.

Pretrained COCO checkpoints. To remove dependence on the training pipeline, the four arms were repeated on torchvision's COCO-pretrained RetinaNet-R50-FPN-v2 [56] and FCOS-R50-FPN [57], without fine-tuning or selection on evaluation data, with calibration on 512 disjoint COCO train2017 images (Table 8). On pretrained RetinaNet, INT8-matched trails FP8-matched by +3.14 [+2.78, +3.57], and excluding the regression head recovers about 82% of this gap (selective minus INT8 +2.58 [+2.33, +2.85]). On pretrained FCOS the clean FP8−INT8 and selective-minus-INT8 contrasts are compatible with zero (+0.05 [−0.22, +0.27]; +0.01 [−0.06, +0.09]). On COCO the interaction is small relative to the level gap (ΔE −0.53 [−0.90, −0.14] AP for RetinaNet and +0.02 [−0.20, +0.28] AP for FCOS), so the deficit is mainly a level shift that corruption does not amplify.

Deployment cost. On one idle host (Supplement S12), INT8-selective is 1.3–6.4% slower than INT8-matched (1.16–1.31 ms per image at batch 1) and 8–11% faster than FP8-matched (1.35–1.49 ms); FP32 references take 4.62–5.09 ms. The regression head accounts for 24.5–25.1% of convolution multiply–accumulate operations (MACs); kernel-level execution precision and the serialized-engine

Table 9
AP by IoU threshold and object size, RetinaNet, J95 basis (single runs). W-only: INT8 head weights, fp16 head activations; A-only: the reverse. Bold: AP_{75} and AP_S of the arms with INT8 head activations, where the deficit concentrates.

Dataset	Arm	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
KITTI	FP8-matched	49.2	77.7	52.8	34.0	48.6	61.3
	INT8-matched	28.2	60.3	**21.6**	**3.0**	23.2	55.4
	INT8-selective	49.5	77.8	52.9	35.7	48.7	61.7
	W-only	49.4	78.0	52.7	35.5	48.6	61.6
	A-only	28.1	60.7	**21.6**	**3.3**	23.2	55.4
VOC	FP8-matched	56.2	81.0	60.9	23.6	37.7	62.2
	INT8-matched	53.2	80.2	**56.9**	**17.9**	33.8	60.3
	INT8-selective	55.8	80.8	60.5	23.7	38.2	61.5
	W-only	55.9	80.9	60.7	23.6	38.3	61.6
	A-only	53.2	80.2	**56.9**	**17.8**	33.7	60.3

weight format were not measured. Five rebuilds of each of seven RetinaNet arms (35 builds) gave bit-different engines; AP, checked for the KITTI matched and selective arms on original-clean and codec-control inputs, was identical.

6. Discussion

6.1. Mechanism of the regression-head deficit

The decomposition and the paired selective arm agree that the regression head carries the deficit, in line with detector PTQ work on regression branches [16]. RetinaNet decodes width and height offsets through an exponential, so a fixed error on the raw output becomes a proportional box error that high-IoU AP penalizes, whereas classification scores pass through a saturating sigmoid. On KITTI the INT8-matched gap to FP8 is 31.2 AP at AP_{75} against 17.4 AP at AP_{50}, and AP_S falls to 3.0 against 34.0, while AP_L drops by only 5.9 AP (Table 9). A-only reproduces this profile and W-only does not. Size-stratified AP also depends on baseline localization, so this profile cannot by itself separate decoding sensitivity from activation quantization.

Activation captures and the Q/DQ structure refine this picture (Supplement S13). Each FPN level has its own per-tensor activation scale at the regression head, so the deficit is not caused by one range shared across levels. The tails, however, are heaviest at the finest levels: median excess kurtosis is 427.5 and 208.1 at P3 and P4 against 2.8 at P7 on KITTI, compared with 20.1 and 19.0 for the classification head, with the same ordering on VOC and a P4 peak on the pretrained COCO graph. Heavier fine-level tails are

consistent with coarser INT8 resolution near the center of the distribution at the levels that predict small objects, which would match the AP_S and AP_{75} collapse. FCOS regression-head kurtosis at P3 and P4 (14.4 and 11.3 on KITTI) is far below the RetinaNet values, consistent with its null. The max-calibration results are compatible with a grid-resolution account, and FP8, whose exponent field provides a wide dynamic range, quantizes the same head without failing (49.48 AP on KITTI). Because max calibration changes clipping and resolution together, the two mechanisms are not isolated, and this is a distributional argument, not a causal test. Why head exclusion helps about eight times more on KITTI than on VOC (21.3 versus 2.6 AP) remains open; differences in object-size distributions are one candidate.

6.2. Decision rules for deploying quantized detectors

The evidence supports five rules.

(i) Calibration preprocessing should match inference byte for byte before any quantization gap is interpreted; on KITTI this alone recovered 21.2 AP in the diagnostic builds.

(ii) Format comparisons under corruption should use the paired, clean-gap-adjusted estimand with absolute corrupted AP, because the FP8–INT8 interaction changed sign between dataset and scale blocks of a single detector family while retention ratios were nearly equal.

(iii) Regression and classification heads should be evaluated as separate quantization regions; when one region carries the deficit, selective precision is the retraining-free repair to try first, since it recovered 99% of the clean-control FP8–INT8 gap on unseen images with lower batch-1 engine latency than FP8.

(iv) When INT8 shows no deficit relative to FP8 and head exclusion brings no gain, as for FCOS, full INT8 should be kept, because selective precision then adds latency without an accuracy gain.

(v) Corruption-augmented calibration should not be expected to transfer; its held-out effect ranged from a significant harm to a +1.66-AP gain, and calibration images should share the deployment codec before gains are credited to corruption coverage.

6.3. Limitations

The paired and calibration experiments use one checkpoint per dataset and model, plus a retrained KITTI RetinaNet checkpoint. Each arm has one engine build and one 512-image calibration draw. The paired experiments use entropy calibration, with max calibration examined only in single builds; testing is otherwise confined to the recorded ModelOpt and TensorRT 11.1 recipes and one RTX 5090; other PTQ methods may distribute quantization error differently. The mechanism analysis covers one anchor-based and one anchor-free family. Intervals cover image sampling only, not training-seed, calibration-draw, or corruption-realization variance. The drive-clustered bootstrap changes KITTI interval widths by a factor of 0.7–3.1 and the significance of 3 of 34 KITTI contrasts, all with an image-bootstrap bound near zero; the RetinaNet deficit, its selective repair, the operand factorial, and the calibration verdicts keep their significance. Holm adjustment covers only the nine YOLO11 blocks; for the 56 FCOS contrasts, the p-value resolution at $B = 2{,}000$ prevents any Holm rejection, so they remain exploratory. The null-seeking selective criterion has no equivalence margin, so an interval containing zero does not prove equivalence. Checkpoints were selected on the KITTI and VOC evaluation partitions; the retrained holdout avoids this for RetinaNet but is split by frame, not by drive. The corruption suite is synthetic, with four types whose severities are not comparable across types, and the calibration experiment tests two folds of these types; on RetinaNet the J95 control differs from original clean images by up to 0.8 AP, enough to move small residuals across significance thresholds; the fold split confounds corruption family with frequency content, held-out JPEG is not fully unseen because all calibration images are JPEG-encoded, and real-world shifts are untested. COCO evidence uses a 2,000-image subset, and the pretrained replication covers one backbone.

7. Conclusion

PDR overcomes the limitations of format-level robustness evaluation identified in Section 1. Its paired, codec-controlled estimand shows that the FP8–INT8 corruption interaction of YOLO11 has no stable sign across datasets and scales, although FP8 has higher absolute AP in every block, so the format label alone does not predict corruption behavior. Its localization stage traces the severe RetinaNet INT8 deficit to the activations consumed by the regression head, with INT8 head weights costing at most about 0.2 AP. Its selective-precision repair outperforms the standard INT8 recipe by 13.85 clean-control AP on images unseen by a retrained checkpoint and by 21.3 AP on KITTI validation, exceeds every calibration-only repair tested, is statistically indistinguishable from FP8 on both KITTI sets (equivalence was not tested; a small residual remains on VOC), and has lower batch-1 engine latency than FP8, without retraining. On FCOS, the same procedure detects no INT8 deficit relative to FP8 and no gain from regression-head exclusion, so plain INT8 is retained. Corruption-aware calibration, by contrast, transfers inconsistently and is not a reliable repair. Future work will apply PDR to transformer detectors, other toolchains, and real-world shifts, and will separate clipping from grid resolution in the regression head.

Data and code availability

The Supplementary Information restates the frozen analysis plan: estimands, condition schedule, arm registry, inference and evaluation constants, calibration schedules, bootstrap schedule, evidence layers, and a reporting checklist. All tables and figures are generated deterministically from frozen ledgers and manifests. The reproducibility package

for this submission is release v3.0.5 (version DOI https://doi.org/10.5281/zenodo.23331336) under concept DOI https://doi.org/10.5281/zenodo.22031663; its SHA-256 manifest binds the manuscript sources, the analysis and pipeline code (including the generator of the repair-comparison table), the activation summaries and quantizer-scale audits, and the metric, input, run, training, and engine-registry records of every reported arm. Raw datasets, trained checkpoints, raw predictions, ONNX graphs, and TensorRT engines are not redistributed; re-running inference requires the documented build pipeline.

Declaration of competing interest

The authors declare no competing interests.

CRediT authorship contribution statement

Dinh Thuan Nguyen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing–original draft. **Lam Phuong Nguyen:** Validation, Data curation, Writing–review and editing. **Vinh Huy Nguyen:** Methodology, Visualization, Writing–review and editing. **Sy Vu Quang:** Validation, Writing–review and editing. **Mohan Rajesh Elara:** Supervision, Writing–review and editing. **Anh Vu Le:** Supervision, Project administration, Writing–review and editing.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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